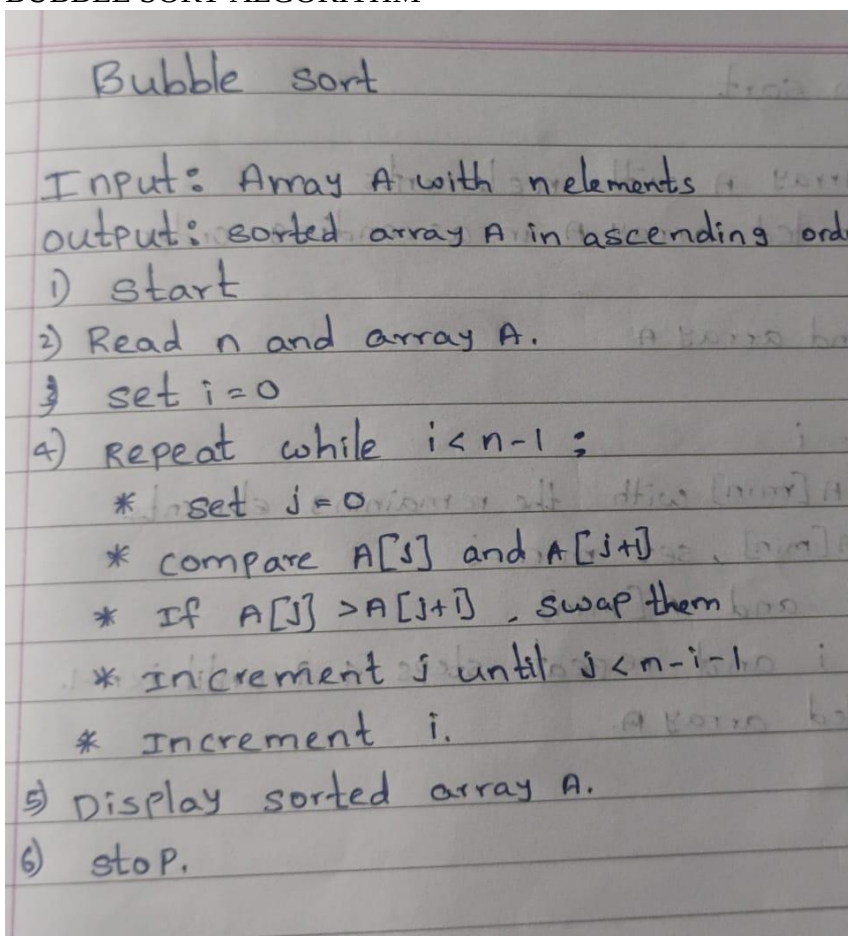




PRACTICAL - 1

AIM : Implementation and Time analysis of sorting algorithms. Bubble sort, Selection sort, Insertion sort, Merge sort and Quicksort

BUBBLE SORT ALGORITHM



PROGRAM:

```
#include <iostream>
using namespace std;
class bubblesort
{
public:
    int size, arr[100], i;
    void arrayinput()
```

```
{
    cout << "Enter array size: ";
    cin >> size;
    cout << "Enter array elements:\n";
    for (i = 0; i < size; i++)
    {
        cin >> arr[i];
    }
}
void printarray()
{
    for (i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}
void bubblelogic()
{
    int j, temp;
    for (i = 0; i < size - 1; i++)
    {
        for (j = 0; j < size - i - 1; j++)
        {
            if (arr[j] > arr[j + 1])
            {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}
};
int main()
{
    bubblesort bs;
    bs.arrayinput();
    cout << "Before sorting: ";
    bs.printarray();
    bs.bubblelogic();
    cout << "After sorting: ";
```



```
bs.printarray();  
return 0;  
}
```

Output

```
Enter array size: 4  
Enter array elements:  
22  
55  
43  
23  
Before sorting: 22 55 43 23  
After sorting: 22 23 43 55
```

TIME COMPLEXITY : Step Count / Frequency Count Method)
best case : Array is already sorted; only one pass is needed

① Bubble sort
best case:-
first pass $\rightarrow n-1$
no swapping
loops stops
 $T(n) = n-1$
Best case = $O(n)$

average case : Elements require multiple comparisons and swaps



```

bubbleSort(arr)
for(i=0; i<n-1; i++)
{
    for(j=0; j<n-i-1; j++)
    {
        if(arr[j] > arr[j+1])
        {
            temp = arr[j];
            arr[j] = arr[j+1];
            arr[j+1] = temp;
        }
    }
}

```

Worst case complexity: $T(n) = (n-1)(n-1) \times 1 \times 1 \times 1$
 $= n^2 - 2n + 1$
 $T(n) = O(n^2)$

Worst case : Array is in reverse order; maximum comparisons and swaps

```

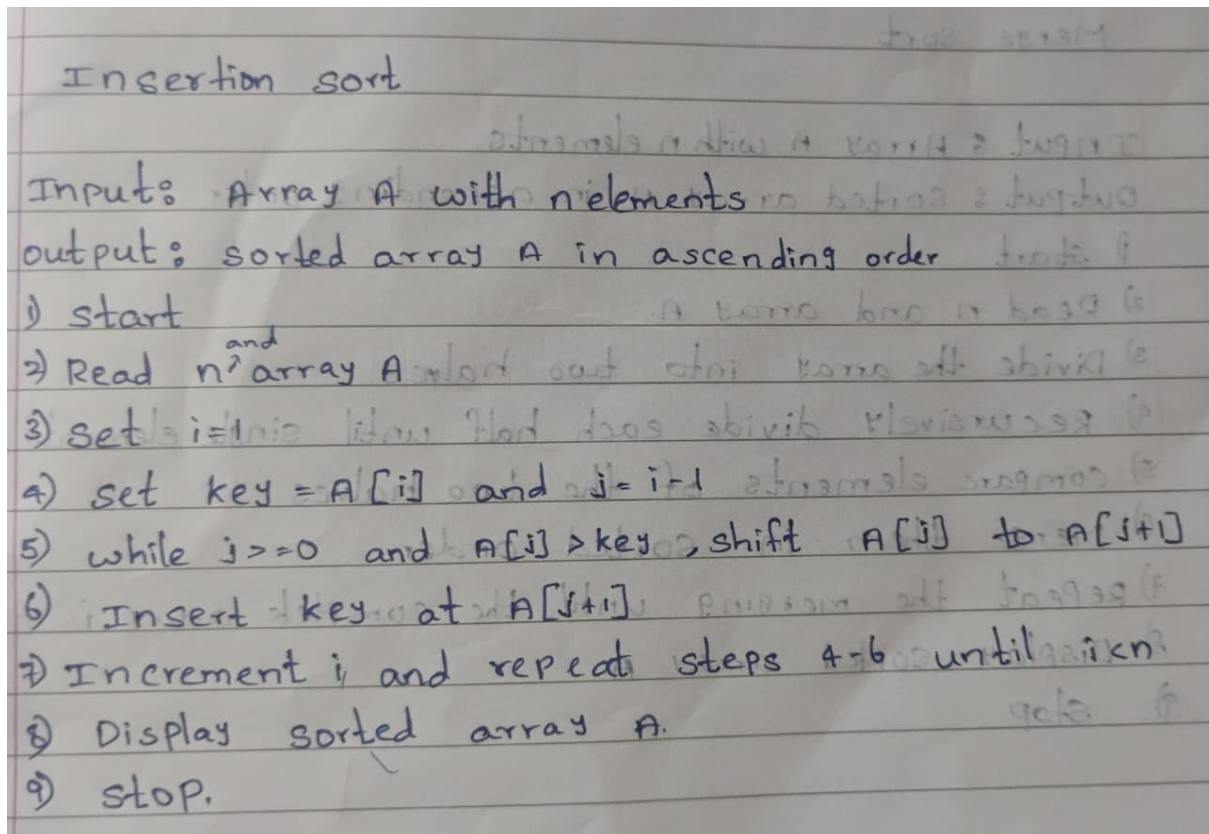
bubbleSort(arr)
for(i=0; i<n-1; i++)
{
    for(j=0; j<n-i-1; j++)
    {
        if(arr[j] > arr[j+1])
        {
            temp = arr[j];
            arr[j] = arr[j+1];
            arr[j+1] = temp;
        }
    }
}

```

Worst case complexity: $T(n) = (n-1)(n-1) \times 1 \times 1 \times 1$
 $= n^2 - 2n + 1$
 $T(n) = O(n^2)$



INSERTION ALGORITHM



PROGRAM:

```
#include <iostream>
using namespace std;

class InsertionSort
{
public:
    int size, arr[100], i, j, temp;

    void arrayinput()
    {
        cout << "Enter array size: ";
        cin >> size;

        cout << "Enter array elements: ";
        for (i = 0; i < size; i++)
```



```
        {
            cin >> arr[i];
        }
    }

void printarray()
{
    for (i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}

void insertionsortlogic()
{
    for (i = 1; i < size; i++)
    {
        temp = arr[i];
        j = i - 1;

        while (j >= 0 && arr[j] > temp)
        {
            arr[j + 1] = arr[j];
            j--;
        }

        arr[j + 1] = temp;
    }
}

};

int main()
{
    InsertionSort is;

    is.arrayinput();

    cout << "Before sorting: ";
    is.printarray();

    is.insertionsortlogic();
```

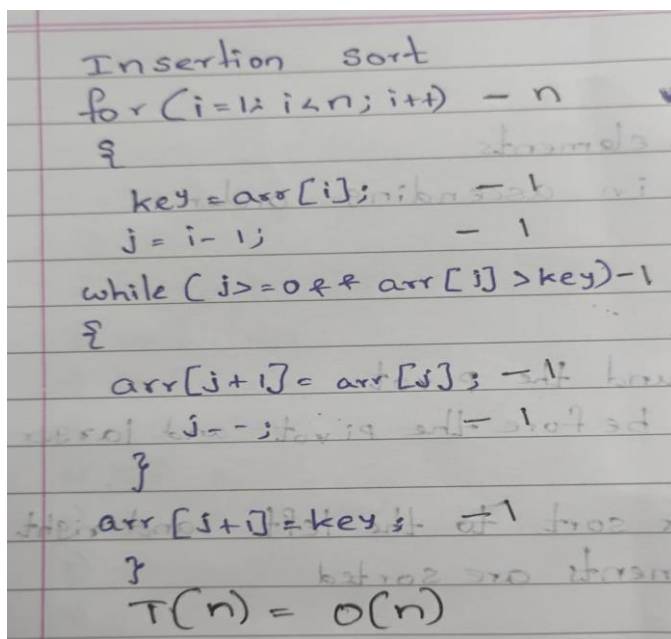
```
cout << "After sorting: ";
is.printarray();

return 0;
}
```

Output

```
Enter array size: 4
Enter array elements: 12
34
76
1
Before sorting: 12 34 76 1
After sorting: 1 12 34 76
```

TIME COMPLEXITY : Step Count / Frequency Count Method)
best case : Array is already sorted; only one comparison per element



```
Insertion sort
for (i = 1; i < n; i++) - n
{
    key = arr[i];
    j = i - 1;
    while (j >= 0 && arr[j] > key) - 1
    {
        arr[j+1] = arr[j];
        j--;
    }
    arr[j+1] = key;
}
T(n) = O(n)
```



average case : Elements are partially unordered, so shifting is required

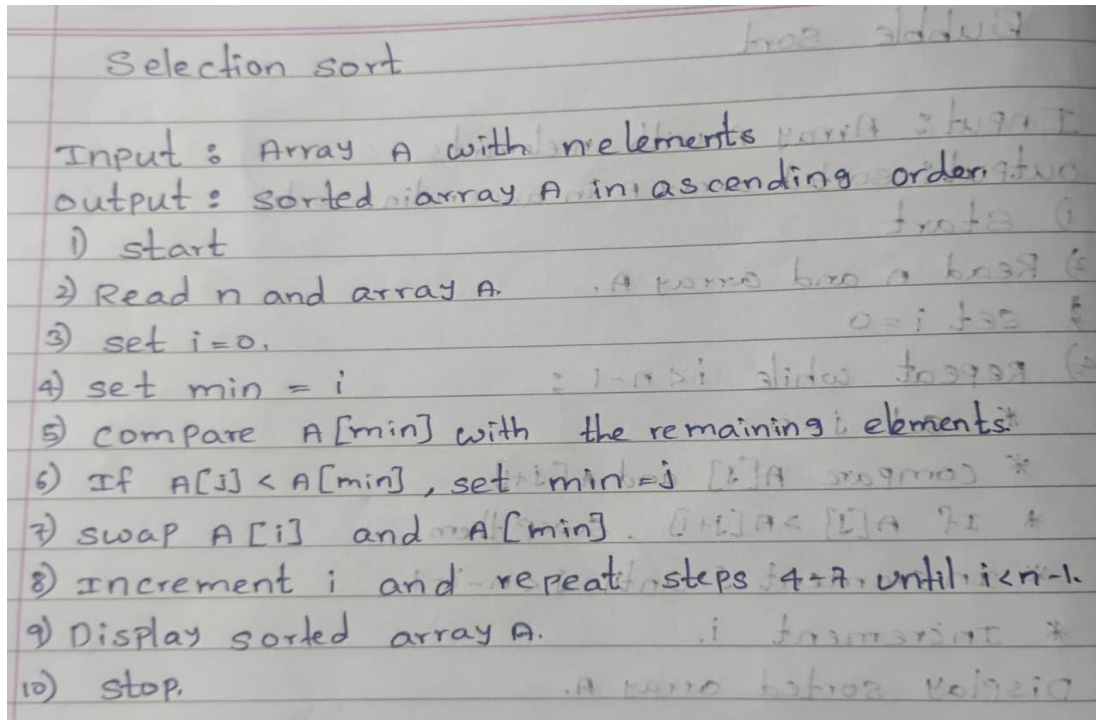
Pass - 1 $\rightarrow n-1$ comparisons
Pass - 2 $\rightarrow n-2$ comparisons
 $\vdots$
 $\therefore T(n) = \frac{n(n-1)}{2}$
ignore constants
Average case $= O(n^2)$

worst case : Array is in reverse order; maximum shifting occurs

The worst case occurs when
Pivot is always smallest (or) largest
 $n + (n-1) + (n-2) + \dots + 1$
 $T(n) = \frac{n(n+1)}{2}$
Worst case $= O(n^2)$



SELECTION SORT ALGORITHM



PROGRAM:

```
#include <iostream>
using namespace std;

class SelectionSort
{
public:
    int size, arr[100], i, j, temp;

    void arrayinput()
    {
        cout << "Enter array size: ";
        cin >> size;

        cout << "Enter array elements: ";
        for (i = 0; i < size; i++)
        {
            cin >> arr[i];
        }
    }
}
```



```
void printarray()
{
    for (i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}

void selectionsortlogic()
{
    for (i = 0; i < size - 1; i++)
    {
        int min = i;

        for (j = i + 1; j < size; j++)
        {
            if (arr[j] < arr[min])
            {
                min = j;
            }
        }

        if (min != i)
        {
            temp = arr[i];
            arr[i] = arr[min];
            arr[min] = temp;
        }
    }
}

};

int main()
{
    SelectionSort ss;

    ss.arrayinput();

    cout << "Before sorting: ";
    ss.printarray();
}
```



```
ss.selectionsortlogic();  
  
cout << "After sorting: ";  
ss.printarray();  
  
return 0;  
}
```

Output

```
Enter array size: 4  
Enter array elements: 23  
46  
11  
98  
Before sorting: 23 46 11 98  
After sorting: 11 23 46 98
```

time complexity : Step Count / Frequency Count Method)
best case : Still compares all remaining elements to find the minimum

```
selection sort  
for (i=0; i<n-1; i++) - n-1  
{  
    min = i; - 1  
    for (j = i+1; j<n; j++) - n  
    {  
        if (arr[j] < arr[min]) - 1  
        {  
            min = j; - 1  
        }  
    }  
    temp = arr[i]; - 1  
    arr[i] = arr[min]; - 1  
    arr[min] = temp; - 1  
}  
T(n) = n(n-1)  
      = n2 - 2n + 1  
      = O(n2)
```



average case :

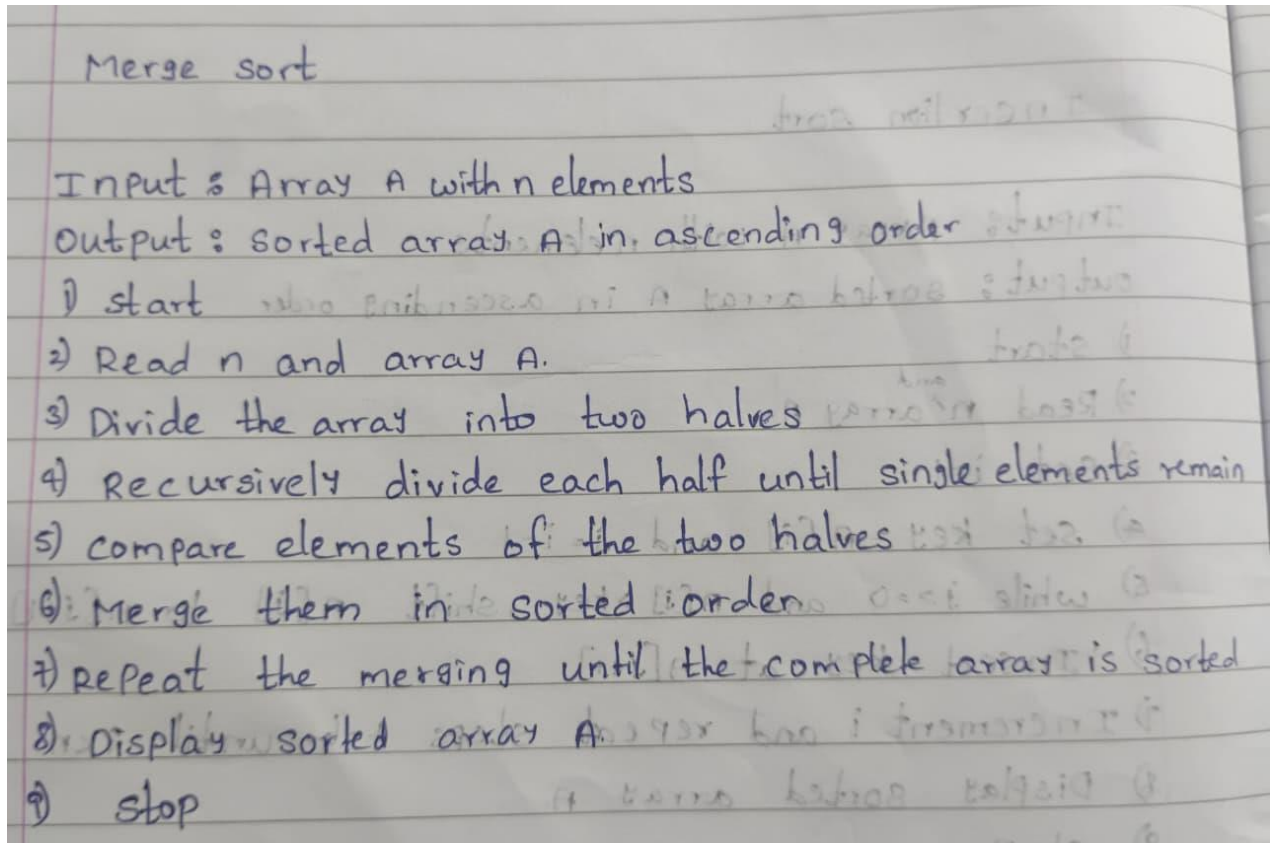
```
selection sort
for (i=0; i<n-1; i++) - n-1
{
    min = i;
    for (j = i+1; j<n; j++) - n
    {
        if (arr[j] < arr[min]) - 1
        {
            min = j;
        }
    }
    temp = arr[i];
    arr[i] = arr[min];
    arr[min] = temp;
}
T(n) = n(n-1)
      = n2 - 2n + 1
      = O(n2)
```

worst case :

```
selection sort
for (i=0; i<n-1; i++) - n-1
{
    min = i;
    for (j = i+1; j<n; j++) - n
    {
        if (arr[j] < arr[min]) - 1
        {
            min = j;
        }
    }
    temp = arr[i];
    arr[i] = arr[min];
    arr[min] = temp;
}
T(n) = n(n-1)
      = n2 - 2n + 1
      = O(n2)
```



MERGE SORT ALGORITHM :



PROGRAM:

```
#include <iostream>
using namespace std;

class mergesort
{
public:
    int size, arr[100], i;

    void arrayinput()
    {
        cout << "Enter array size: ";
        cin >> size;

        cout << "Enter array elements:\n";
        for (i = 0; i < size; i++)
        {
```

```
        cin >> arr[i];
    }
}

void printarray()
{
    for (i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}

void merge(int low, int mid, int high)
{
    int temp[100];
    int i = low, j = mid + 1, k = low;

    while (i <= mid && j <= high)
    {
        if (arr[i] <= arr[j])
        {
            temp[k] = arr[i];
            i++;
        }
        else
        {
            temp[k] = arr[j];
            j++;
        }
        k++;
    }

    while (i <= mid)
    {
        temp[k] = arr[i];
        i++;
        k++;
    }

    while (j <= high)
    {
```

```
        temp[k] = arr[j];
        j++;
        k++;
    }

    for (i = low; i <= high; i++)
    {
        arr[i] = temp[i];
    }
}

void mergellogic(int low, int high)
{
    if (low < high)
    {
        int mid = (low + high) / 2;

        mergellogic(low, mid);
        mergellogic(mid + 1, high);

        merge(low, mid, high);
    }
}

};

int main()
{
    mergesort ms;

    ms.arrayinput();

    cout << "Before sorting: ";
    ms.printarray();

    ms.mergellogic(0, ms.size - 1);

    cout << "After sorting: ";
    ms.printarray();

    return 0;
}
```

Output

```
Enter array size: 4
Enter array elements:
12
45
23
11
Before sorting: 12 45 23 11
After sorting: 11 12 23 45
```

time complexity : Step Count / Frequency Count Method)

best case : Array is divided into $\log n$ levels and merging takes $O(n)$



```

Merge sort
void mergesort (int arr[], int low, int high)
{
    if (low < high)
    {
        mid = (low + high) / 2;
        mergesort (arr, low, mid);
        mergesort (arr, mid+1, high);
        merge (arr, low, mid, high);
    }
}

void merge (int arr[], int low, int mid, int high)
{
    i = low;
    j = mid + 1;
    k = low;
    while (i <= mid && j <= high)
    {
        if (arr[i] < arr[j])
        {
            temp[k] = arr[i];
            i++;
        }
        else
        {
            temp[k] = arr[j];
            j++;
        }
        k++;
    }
    while (i <= mid)
    {
        temp[k] = arr[i];
        i++;
        k++;
    }
    while (j <= high)
    {
        temp[k] = arr[j];
        j++;
        k++;
    }
    for (i = low; i <= high; i++)
    {
        arr[i] = temp[i];
    }
}

```

Recurrence Relation

$$T(n) = 2T(n/2) + O(n)$$

expand

$$T(n) = 2[2T(n/4) + n/2] + n$$

$$T(n) = 4T(n/4) + 2n$$

again

$$T(n) = 8T(n/8) + 3n$$

$$= 16T(n/16) + 4n$$

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

$$n = 2^k$$

add log on both sides

$$T(n) = 2 \log^n(1) + \log n(n)$$

$$T(n) = n \log^2 + n \log n$$

$$T(n) = n(1) + n \log n$$

$$\boxed{T(n) = n \log n}$$

average case :

Recurrence Relation

$$T(n) = 2T(n/2) + O(n)$$

expand

$$T(n) = 2[2T(n/4) + n/2] + n$$

$$T(n) = 4T(n/4) + 2n$$

again

$$T(n) = 8T(n/8) + 3n$$

$$= 16T(n/16) + 4n$$

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

$$n = 2^k$$

add log on both sides

$$T(n) = 2 \log^n(1) + \log n(n)$$

$$T(n) = n \log^2 + n \log n$$

$$T(n) = n(1) + n \log n$$

$$\boxed{T(n) = n \log n}$$

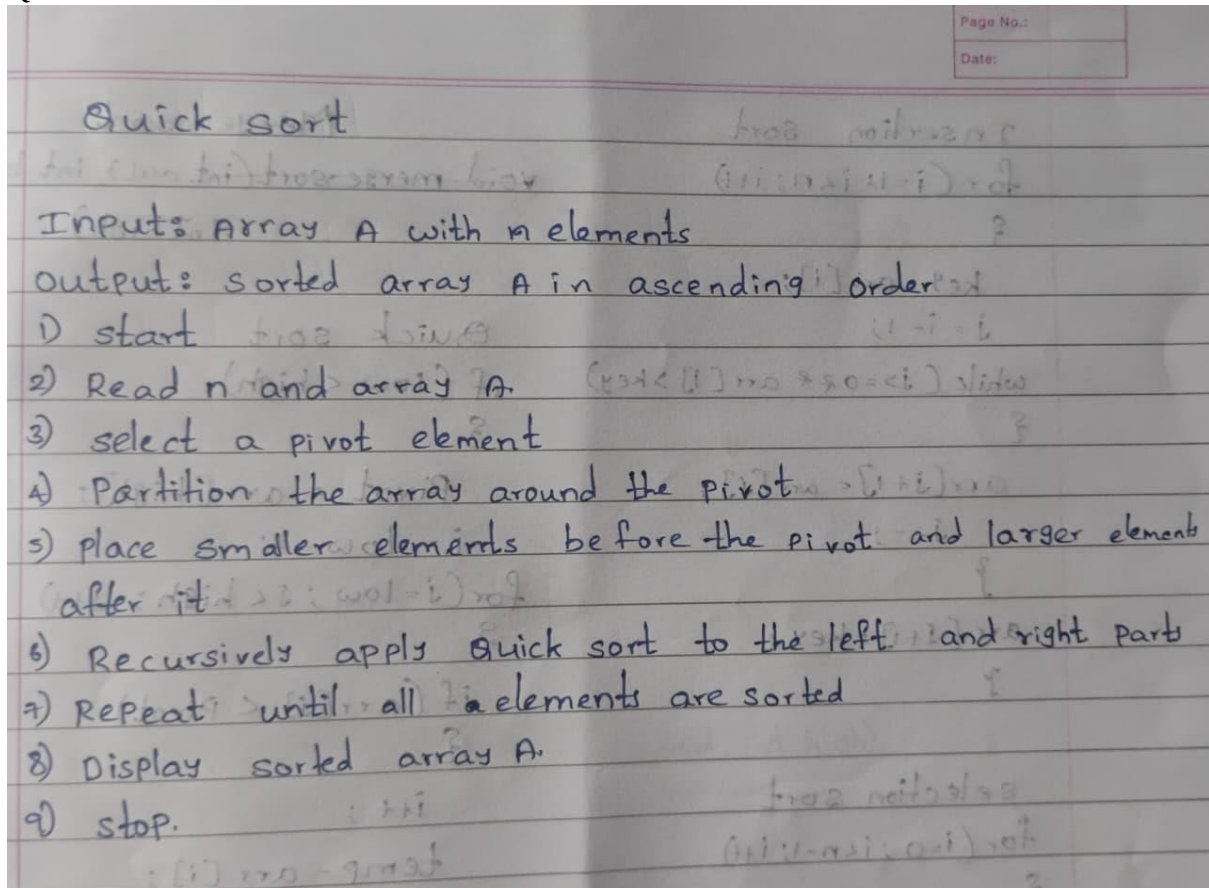


worst case :

Recurrence Relation
 $T(n) = 2T(n/2) + O(n)$
expand
 $T(n) = 2[2T(n/4) + n/2] + n$
 $T(n) = 4T(n/4) + 2n$
again
 $T(n) = 8T(n/8) + 3n$
 $= 16T(n/16) + 4n$
 $T(n) = 2^k T(n/2^k) + kn$
 $n = 2^k$
add log on both sides
 $T(n) = 2 \log n (1) + \log n (n)$
 $T(n) = n \log^2 + n \log n$
 $T(n) = n(1) + n \log n$
 $T(n) = n \log n$



QUICK SORT ALGORITHM:



PROGRAM :

```
#include <iostream>
#include <utility>
using namespace std;

class QuickSort
{
public:
    int size, arr[100];

    void arrayinput()
    {
        cout << "Enter array size: ";
        cin >> size;

        cout << "Enter array elements: ";
```



```
    for (int i = 0; i < size; i++)
    {
        cin >> arr[i];
    }
}

void printarray()
{
    for (int i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}

int partition(int lb, int ub)
{
    int pivot = arr[lb];
    int start = lb;
    int end = ub;

    while (start < end)
    {
        while (start <= ub && arr[start] <= pivot)
            start++;

        while (arr[end] > pivot)
            end--;

        if (start < end)
            swap(arr[start], arr[end]);
    }

    swap(arr[lb], arr[end]);
    return end;
}

void quicksortlogic(int lb, int ub)
{
    if (lb < ub)
    {
        int loc = partition(lb, ub);
```

```
        quicksortlogic(lb, loc - 1);
        quicksortlogic(loc + 1, ub);
    }
}

void sort()
{
    quicksortlogic(0, size - 1);
}
};

int main()
{
    QuickSort qs;

    qs.arrayinput();

    cout << "Before sorting: ";
    qs.printarray();

    qs.sort();

    cout << "After sorting: ";
    qs.printarray();

    return 0;
}
```

Output

```
Enter array size: 4
Enter array elements: 23
43
55
12
Before sorting: 23 43 55 12
After sorting: 12 23 43 55
```

TIME COMPLEXITY : Step Count / Frequency Count Method)
best case : Pivot divides the array into nearly equal parts



```
Quick sort
if (low < high) {
    Pivot = arr[high];
    i = low - 1;
    for (j = low; j < high; j++) {
        if (arr[j] < Pivot) {
            i++;
            temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }
    temp = arr[i+1];
    arr[i+1] = arr[high];
    arr[high] = temp;
    P = i+1;
    quickSort(arr, low, P-1);
    quickSort(arr, P+1, high);
}
```



Recurrence Relation

$$T(n) = 2T(n/2) + O(n)$$

expand

$$T(n) = 2[2T(n/4) + n/2] + n$$

$$T(n) = 4T(n/4) + 2n$$

again

$$T(n) = 8T(n/8) + 3n$$

$$= 16T(n/16) + 4n$$

$$T(n) = 2^k T(n/2^k) + kn$$

$$n = 2^k$$

add log on both sides

$$T(n) = 2 \log n (1) + \log n (n)$$

$$T(n) = n \log^2 + n \log n$$

$$T(n) = n(1) + n \log n$$

$$\boxed{T(n) = n \log n}$$

average case : Pivot usually creates reasonably balanced partitions

Recurrence Relation

$$T(n) = 2T(n/2) + O(n)$$

expand

$$T(n) = 2[2T(n/4) + n/2] + n$$

$$T(n) = 4T(n/4) + 2n$$

again

$$T(n) = 8T(n/8) + 3n$$

$$= 16T(n/16) + 4n$$

$$T(n) = 2^k T(n/2^k) + kn$$

$$n = 2^k$$

add log on both sides

$$T(n) = 2 \log n (1) + \log n (n)$$

$$T(n) = n \log^2 + n \log n$$

$$T(n) = n(1) + n \log n$$

$$\boxed{T(n) = n \log n}$$



worst case : Pivot is always the smallest or largest element

The worst case occurs when
Pivot is always smallest (or) largest

$$n + (n-1) + (n-2) + \dots + 1$$
$$T(n) = \frac{n(n+1)}{2}$$
$$\text{Worst case} = O(n^2)$$